Q1)

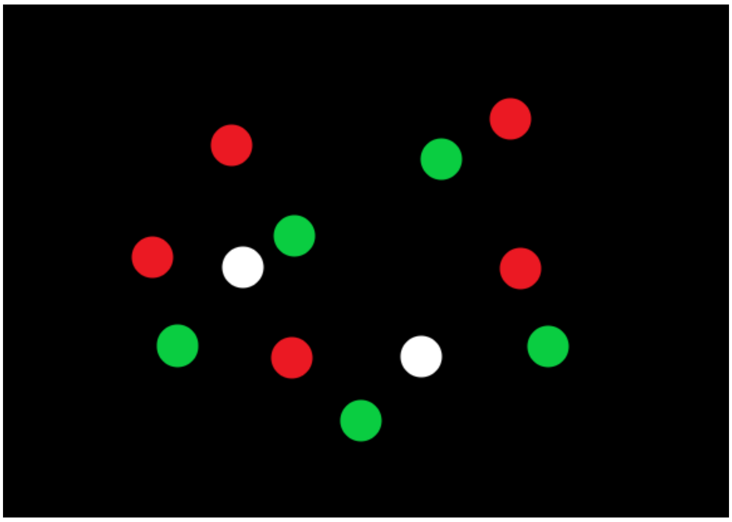
A)

1. F, Mean Shift
2. F , $\alpha > 0$
3. مش علينا
4. F, Fast**er** R-CNN
5. مش علينا
6. F, Feature Detector
7. مش علينا
8. T
9. F , Image rotation
10. F, Same Label
11. مش علينا
12. F, Low mAP

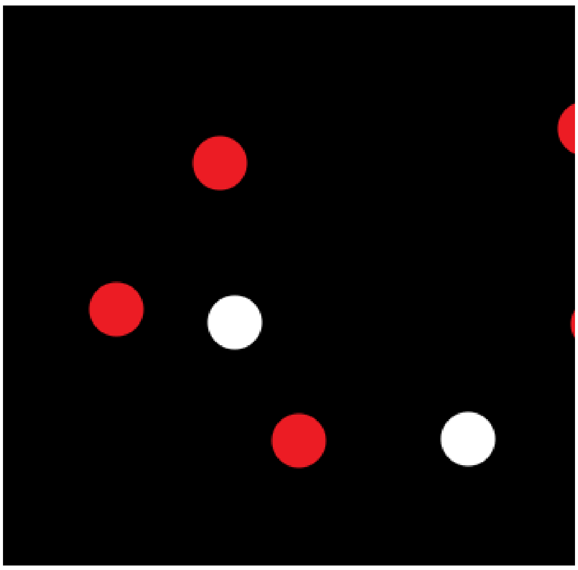
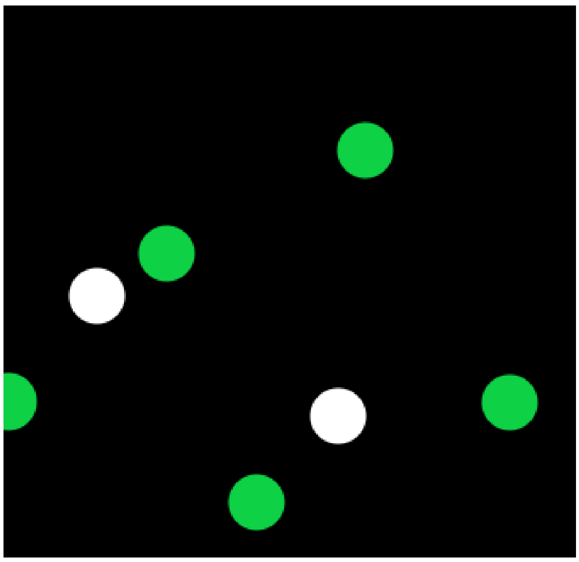
B) مش علينا

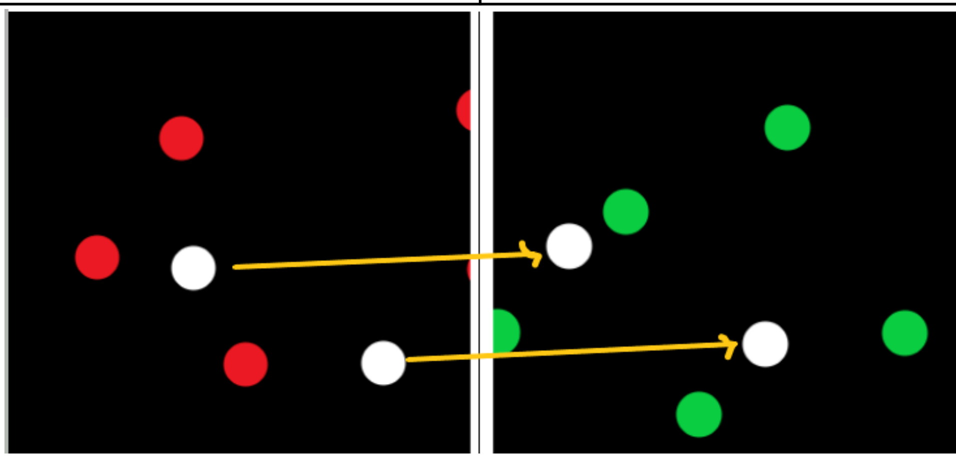
Q2)

A)



let C_R be the camera that sees Red and C_G be the Camera that sees Green

C_R view:	C_G view:
	



i.

C_R : sees 5 red points and 2 White points

C_G : sees 5 green points and 2 White points

ii.

Yes, Since we have 2 points which are visible to both cameras (white points) so we can get 2 matches which is sufficient for estimating a homography matrix of translation that contains only 2 unknowns

iii.

The RANSAC algorithm

Algorithm:

1. **Sample** (randomly) the number of points required to fit the model.
2. **Solve** for model parameters using sample.
3. **Score** by the fraction of ***inliers*** within a preset threshold of the model.
4. **Repeat** 1-3 until the best model is found with high confidence.

Parameters:

- Initial number of points s
  - Typically minimum number needed to fit the model
- Distance threshold T
  - Choose T so probability for inlier is high (e.g. 0.95)
- Number of iterations N
  - Choose N so that, with probability p, at least one random sample set is free from outliers (e.g. p = 0.99)

B)

I.

$$E(u,v) = \sum_{(x,y)\in W} [I(x+u, y+v) - I(x,y)]^2$$

II.

$$M = \begin{bmatrix} \sum I_x^2 & \sum I_x I_y \\ \sum I_x I_y & \sum I_y^2 \end{bmatrix}$$

Good corners are areas where both $\sum I_x^{\ 2}$ and $\sum I_y^{\ 2}$ are large

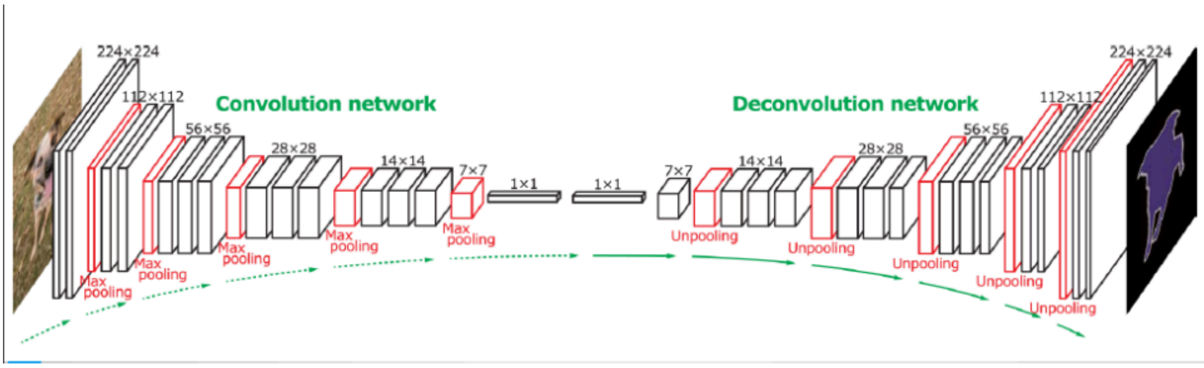
III.

1. Compute derivatives at each pixel
2. Compute second moment matrix M in a Gaussian window around each pixel
3. Compute corner response function R
4. Threshold R
5. Find local maxima of response function (non-maximum suppression)

Q3)

A)

A Deconvolution architecture can be used for such task by first passing the image through convolution layers for feature extraction then deconvolution for predicting the semantic map



B) مش علينا

Q4) مش علينا